BIOL-1 Practice Test 02 — Answer Sheet

Modules 5-6: Membranes and Metabolism

Name:		**Date:**		**Score:**	

✂️ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice and Part B: Fill in the Blank answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice

1.		**6.**		**11.**	
2.		**7.**		**12.**	
3.		**8.**		**13.**	
4.		**9.**			
5.		**10.**			

Part B: Fill in the Blank

1.		**6.**	
2.		**7.**	
3.		**8.**	
4.		**9.**	
5.			

BIOL-1 Practice Test 02 — Part C: Short Answer

Free response. Write clearly in the lined spaces.

1. Explain the difference between passive transport and active transport. Give one example of each.
2. Describe what happens to a cell placed in a hypertonic solution. How would this differ for a plant cell versus an animal cell?
3. Explain how enzymes speed up chemical reactions and why they are specific to their substrates.
4. Describe the structure of ATP and explain how it stores and releases energy for the cell.

Response 1 — Question Number: ______

Response 2 — Question Number: ______

Response 3 — Question Number: ______

Response 4 — Question Number: ______

BIOL-1 Practice Test 02 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice

Choose the best answer for each question.

Module 5: Membranes

1. The cell membrane is best described by the:

A) Fluid mosaic model
B) Rigid barrier model
C) Lock and key model
D) Simple lipid model

2. The cell membrane is primarily composed of:

A) Proteins only
B) Phospholipid bilayer
C) Carbohydrates
D) Nucleic acids

3. Phospholipids spontaneously form a bilayer in water because:

A) They are charged molecules
B) They have hydrophilic heads and hydrophobic tails
C) They are completely hydrophobic
D) They are all the same size

4. Cholesterol in the cell membrane functions to:

A) Produce ATP
B) Store genetic information
C) Help maintain membrane fluidity
D) Transport molecules across the membrane

5. Which transport process requires ATP?

A) Diffusion
B) Facilitated diffusion
C) Osmosis
D) Active transport

6. When a cell is placed in a hypotonic solution, water will:

A) Move into the cell
B) Not move
C) Move in both directions equally
D) Move out of the cell

7. A cell in an isotonic solution will experience:

A) Net water movement into the cell
B) Cell lysis
C) No net water movement
D) Net water movement out of the cell

Module 6: Metabolism

8. Enzymes function by:

A) Providing energy for reactions
B) Decreasing activation energy
C) Increasing activation energy
D) Changing the equilibrium of reactions

9. The molecule that stores and transfers energy in all cells is:

A) Glucose
B) DNA
C) ATP
D) NADH

10. The part of an enzyme where the substrate binds is called the:

A) Active site
B) Product site
C) Inhibitor site
D) Allosteric site

11. When an enzyme loses its shape due to heat or pH changes, this is called:

A) Activation
B) Catalysis
C) Inhibition
D) Denaturation

12. ATP releases energy when:

A) A phosphate bond is broken (hydrolysis)
B) It binds to DNA
C) It is stored in the cell
D) A phosphate group is added

13. Which statement about enzymes is TRUE?

A) Enzymes are made of carbohydrates
B) Enzymes increase the activation energy
C) Enzymes are used up in reactions
D) Enzymes can be reused

Part B: Fill in the Blank

Write the correct term in the blank.

14. The phospholipid bilayer is described as a ________ mosaic model.

15. The process by which water moves across a membrane is called ________.

16. Transport that requires ATP is called ________ transport.

17. A solution with a lower solute concentration than the cell is called ________.

18. The specific reactant that an enzyme acts on is called the ________.

19. Energy of motion is referred to as ________ energy.

20. ________ inhibition occurs when a molecule binds to the active site and blocks the substrate.

21. The sum of all chemical reactions in an organism is called ________.

22. Reactions that release energy by breaking down complex molecules are called ________ reactions.
